

WINSTON FIELD, TX, USA

WMO: 722122

Lat: 32.693N

Lon: 100.951W

Elev: 741

StdP: 92.74

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03050

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.6	(c) -5.2	(d) -19.9	(e) 0.7	(f) 12.2	(g) -16.7	(h) 0.9	(i) 10.0	(j) 13.6	(k) 12.3	(l) 12.1	(m) 10.2	(n) 3.6	(o) 20	(p) 0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	37.8	19.9	36.5	20.3	35.3	20.5	23.9	31.9	23.3	31.0	22.8	30.3	5.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 22.1	(b) 18.4	(c) 26.3	(d) 21.3	(e) 17.5	(f) 25.7	(g) 20.9	(h) 17.0	(i) 25.5	(j) 75.6	(k) 32.0	(l) 73.2	(m) 31.3	(n) 71.0	(o) 30.1	(p) 27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.5	10.1	8.8	-10.9	39.6	2.5	2.0	-12.7	41.0	-14.1	42.2	-15.5	43.3	-17.3	44.7
			-11.8	25.1	2.4	1.0	-13.5	25.9	-14.9	26.5	-16.2	27.0	-17.9	27.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	6.5	8.8	13.7	17.9	22.1	26.7	27.8	27.6	23.7	17.8	11.7	6.8
	DBStd	8.93	5.15	5.98	5.25	4.92	4.75	2.89	2.40	2.62	3.45	4.84	4.99	5.22
	HDD10.0	416	130	86	25	5	1	0	0	0	0	6	39	124
	HDD18.3	1508	367	269	157	66	20	0	0	0	3	66	203	357
	CDD10.0	3212	22	51	141	242	377	502	553	546	412	249	91	26
	CDD18.3	1261	0	2	14	52	138	252	294	287	165	51	4	0
	CDH23.3	14517	9	59	227	672	1531	2922	3576	3390	1570	483	68	11
	CDH26.7	7261	0	11	61	271	741	1546	1936	1839	689	157	10	0
Wind	WSAvg	4.3	4.3	4.6	5.0	5.4	4.8	4.5	3.8	3.6	3.6	4.0	4.0	4.1
Precipitation	PrecAvg	563	18	25	31	39	75	78	49	63	75	60	30	19
	PrecMax	1087	62	114	154	150	225	198	160	424	328	170	206	111
	PrecMin	307	0	0	0	0	2	0	0	0	0	0	0	0
	PrecStd	159	17	24	32	41	60	50	46	79	68	49	40	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.7	28.3	31.6	35.0	38.0	39.8	38.6	39.0	36.3	32.6	27.9	25.0
		MCWB	11.3	13.1	13.8	13.9	16.4	18.6	21.0	20.3	19.9	19.4	14.8	11.3
	2%	DB	21.5	25.1	28.2	32.4	35.5	37.6	37.3	37.6	34.6	30.9	25.2	21.2
		MCWB	9.8	11.5	13.1	14.7	17.4	20.0	20.6	20.8	20.2	18.7	14.3	11.0
	5%	DB	18.7	22.3	26.0	30.5	33.5	35.7	35.9	36.1	32.6	28.0	23.0	18.0
		MCWB	8.7	11.2	13.3	15.2	18.4	20.7	21.0	20.9	20.2	17.9	13.4	10.0
	10%	DB	15.4	19.3	23.6	27.3	31.5	34.3	34.9	34.9	31.2	26.1	20.6	15.4
		MCWB	8.0	10.1	12.8	15.0	18.5	20.9	21.2	21.1	20.2	17.0	12.6	8.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.2	16.3	19.0	20.6	23.0	24.9	24.7	24.6	23.9	22.4	18.0	15.7
		MCDB	17.3	21.8	23.5	26.7	29.9	32.9	32.6	32.1	30.8	28.1	23.0	18.5
	2%	WB	12.1	14.5	17.7	19.5	22.2	23.7	23.7	23.6	22.7	21.3	16.6	13.4
		MCDB	17.3	20.5	23.0	24.7	28.7	31.8	31.7	31.6	28.8	26.6	21.6	17.1
	5%	WB	10.3	12.7	16.4	18.5	21.5	23.0	23.0	23.0	22.0	20.3	15.2	11.2
		MCDB	17.4	19.1	21.3	23.9	27.7	30.7	30.9	31.1	27.9	25.3	19.7	16.6
	10%	WB	8.5	10.9	14.9	17.2	20.7	22.4	22.5	22.4	21.4	19.2	13.8	9.3
		MCDB	14.7	18.1	20.5	23.3	26.7	29.9	30.2	30.3	27.1	23.9	18.9	15.5
Mean Daily Temperature Range	5% DB	MDBR	13.8	14.1	14.3	14.6	12.9	12.6	12.1	12.1	11.6	12.8	13.5	12.6
		MCDBR	19.1	19.3	17.7	18.1	16.1	14.8	13.7	13.9	14.0	15.3	16.6	17.6
		MCWBR	9.7	9.4	7.9	7.6	6.7	5.1	4.5	4.4	4.7	6.9	8.0	9.4
	5% WB	MCDBR	16.4	15.9	13.8	13.4	13.2	12.9	11.9	12.2	11.0	12.4	12.9	14.4
		MCWBR	9.4	9.6	7.6	7.6	6.1	5.0	4.4	4.3	4.1	6.6	7.8	8.9
Clear-Sky Solar Irradiance	taub	0.276	0.287	0.315	0.344	0.357	0.394	0.399	0.388	0.360	0.337	0.304	0.284	
	taud	2.563	2.538	2.470	2.393	2.430	2.359	2.409	2.407	2.439	2.486	2.509	2.542	
	Ebn at Noon	958	977	966	944	928	889	883	891	905	906	913	926	
	Edh at Noon	80	91	105	119	116	124	118	116	108	95	84	78	
All-Sky Solar Radiation	RadAvg	3.32	4.06	5.30	6.50	6.86	7.37	7.07	6.54	5.41	4.36	3.48	2.91	
	RadStd	0.34	0.48	0.37	0.44	0.61	0.45	0.43	0.35	0.38	0.57	0.39	0.29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (15 neighbors)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+121	+83

Nomenclature: See separate page